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System and Method for Extracting Hintsights for Assortment Planning

Abstract

A system and method are disclosed for extracting hintsights for assortment planning. The system provides for classifying each of one or more products in a product display area of a retail entity as high-performers or low-performers according to selected metrics. The system further provides for identifying a strength of each individual product attribute associated with each of the products, deriving a multi-combination strength of combinations of product attributes using the identified strength of each individual product attribute, and generating recommendations for product combinations based on the derived multi-combination strength of combinations of product attributes and the selected metrics.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/878,798, filed Aug. 1, 2022, entitled “System and Method for Extracting Hindsights for Assortment Planning,” which claims the benefit under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/227,553, filed Jul. 30, 2021, entitled “System and Method for Extracting Hindsights for Assortment Planning.” U.S. patent application Ser. No. 17/878,798 and U.S. Provisional Application No. 63/227,553 are assigned to the assignee of the present application.

TECHNICAL FIELD

[0002] The present disclosure relates generally to data processing, and more particularly to processing data relating to retail assortment planning.

BACKGROUND

[0003] Assortment planning in a supply chain typically involves determining what products to stock in a particular retail location. Such analysis is typically based on previous sales data, in combination with the business knowledge and intuition of a supply chain planner. However, the results of such a process may be inaccurate due to inherent uncertainties in the supply chain planner's intuition. Some supply chain systems instead attempt to model demand for products at the retail location, using supply chain machine learning systems to generate one or more models to model the flow of materials and the demand for products across supply chain networks and the individual entities of the supply chain. However, systems that attempt to derive useful insights regarding the potential sales of one or more product attribute combinations using machine learning may be extremely computationally intensive and may fail to locate insights within feasible timeframes, both of which are undesirable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] A more complete understanding of the present invention may be derived by referring to the detailed description when considered in connection with the following illustrative figures. In the figures, like reference numbers refer to like elements or acts throughout the figures.

[0005] FIG. 1 illustrates an exemplary supply chain network according to a first embodiment;

[0006] FIG. 2 illustrates the assortment planner of FIG. 1 in greater detail in accordance with an embodiment;

[0007] FIG. 3 illustrates an exemplary method of the assortment planner extracting hindsights from assortment planning data, according to an embodiment;

[0008] FIG. 4 illustrates an individual attribute strength data graphical user interface display, according to an embodiment;

[0009] FIG. 5 illustrates a multi-attribute strength data graphical user interface display, according to an embodiment;

[0010] FIG. 6 illustrates a strength, weakness, opportunity and threat analysis graphical user interface display, according to an embodiment;

[0011] FIG. 7 illustrates a second method by which a hindsight engine may estimate the multi-combination strength of attributes, according to embodiments; and

[0012] FIG. 8 illustrates an insight analysis graphical user interface display. according to an

embodiment.

DETAILED DESCRIPTION

[0013] Aspects and applications of the invention presented herein are described below in the drawings and detailed description of the invention. Unless specifically noted, it is intended that the words and phrases in the specification and the claims be given their plain, ordinary, and accustomed meaning to those of ordinary skill in the applicable arts.

[0014] In the following description, and for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the various aspects of the invention. It will be understood, however, by those skilled in the relevant arts, that the present invention may be practiced without these specific details. In other instances, known structures and devices are shown or discussed more generally in order to avoid obscuring the invention. In many cases, a description of the operation is sufficient to enable one to implement the various forms of the invention, particularly when the operation is to be implemented in software. It should be noted that there are many different and alternative configurations, devices and technologies to which the disclosed inventions may be applied. The full scope of the inventions is not limited to the examples that are described below.

[0015] As described below, embodiments of the following disclosure provide a system and method to analyze assortment planning and potential combinations of product attributes, determine hindsight and useful information from the analysis, and to recommend potential combinations of product attributes (such as, for example, preparing additional red T-shirts with a particular design logo and collar style) according to one or more selected metrics. Embodiments comprise an assortment planner that, among other actions, aggregates product-store combinations of attributes and classifies high-performing and low-performing combinations. Embodiments identify attribute strength in both single-attribute and multi-attribute contexts, and filter attributes according to attribute strengths. Embodiments utilize the attribute strengths to generate recommendations for product-store and/or product/attribute combinations based on the one or more metrics.

[0016] Embodiments of the present invention enable retailers and other supply chain entities to utilize both frequentist and Bayesian analysis techniques to conduct hindsight analysis and to identify potentially high-performing product attribute combinations in both supervised and unsupervised contexts. Embodiments search for potentially useful cross-combinations of products and attributes using computationally efficient methods and procedures, and generate directly-applicable recommendations and hindsight quickly and with reduced computational runtime requirements. Embodiments allow supply chain planners to make use of such recommendations to perform assortment planning for one or more retail locations which is based on an attribute analysis, rather than utilizing human intuition.

[0017] FIG. 1 illustrates an exemplary supply chain network **100** according to a first embodiment. Supply chain network **100** comprises assortment planner **110**, inventory system **120**, transportation network **130**, one or more supply chain entities **140**, computer **150**, network **160**, and communication links **162-170**. Although a single assortment planner **110**, inventory system **120**, transportation network **130**, one or more supply chain entities **140**, a single computer **150**, and a single network **160** are illustrated and described, embodiments contemplate any number of assortment planners **110**, inventory systems **120**, transportation networks **130**, supply chain entities **140**, computers **150**, or networks **160**, according to particular needs.

[0018] In one embodiment, assortment planner **110** comprises server **112** and database **114**. As described in more detail below, assortment planner **110** comprises one or more modules to, for example, generate hindsight and recommendations for product attributes based on identified attribute strength in single-attribute and multi-attribute contexts.

[0019] Inventory system **120** comprises server **122** and database **124**. Server **122** is configured to receive and transmit inventory data, including item identifiers, pricing data, attribute data, inventory levels, and other like data about one or more items or products at one or more locations

in supply chain network **100**. Server **122** stores and retrieves inventory data from database **124** or from one or more locations in supply chain network **100**.

[0020] According to embodiments, database **124** includes current or projected inventory quantities or states, order rules, or explanatory variables. For example, database **124** may comprise the current level of inventory for each item at one or more stocking points across supply chain network **100**. In addition, database **124** may comprise order rules that describe one or more rules or limits on setting an inventory policy, including, but not limited to, a minimum order quantity, a maximum order quantity, a discount, and a step-size order quantity, and batch quantity rules. According to some embodiments, database **124** may comprise explanatory variables that describe the data relating to specific past, current, or future indicators and the data of promotions, seasonality, special events (such as sporting events), weather, and the like. According to some embodiments, assortment planner **110** accesses and stores inventory data in database **124**, which may be used by assortment planner **110** to generate one or more planograms, set inventory levels at one or more stocking points, initiate manufacturing of one or more components, or the like. In addition, or as an alternative, the inventory data of database **124** may be updated by receiving current item quantities, mappings, or locations from inventory system **120**, transportation network **130**, and/or one or more supply chain entities **140**.

[0021] Transportation network **130** comprises server **132** and database **134**. According to embodiments, transportation network **130** directs one or more transportation vehicles **136** to ship one or more items between one or more supply chain entities **140**, based, at least in part, on the planograms and instruction sets generated by assortment planner **110**. Transportation vehicles **136** comprise, for example, any number of trucks, cars, vans, boats, airplanes, unmanned aerial vehicles (UAVs), cranes, robotic machinery, or the like. Transportation vehicles **136** may comprise radio, satellite, or other communication systems that communicates location information (such as, for example, geographic coordinates, distance from a location, global positioning satellite (GPS) information, or the like) with assortment planner **110**, inventory system **120**, transportation network **130**, and/or one or more supply chain entities **140** to identify the location of transportation vehicle **136** and the location of any inventory or shipment located on transportation vehicle **136**. In addition to the supply chain models, the number of items shipped by transportation vehicles **136** in transportation network **130** may also be based, at least in part, on the number of items currently in stock at one or more supply chain entities **140**, the number of items currently in transit in transportation network **130**, a forecasted demand, a supply chain disruption, and the like.

[0022] As shown in FIG. **1**, supply chain network **100** operates on one or more computers **150** that are integral to or separate from the hardware and/or software that support assortment planner **110**, inventory system **120**, transportation network **130**, and one or more supply chain entities **140**. Supply chain network **100** comprising assortment planner **110**, inventory system **120**, transportation network **130**, and one or more supply chain entities **140** may operate on one or more computers **150** that are integral to or separate from the hardware and/or software that support assortment planner **110**, inventory system **120**, transportation network **130**, and one or more supply chain entities **140**. One or more computers **150** may include any suitable input device **152**, such as a keypad, mouse, touch screen, microphone, or other device to input information. One or more computers **150** may also include any suitable output device **154**, such as, for example, a computer monitor, that may convey information associated with the operation of supply chain network **100**, including digital or analog data, visual information, or audio information. Computer **150** may include fixed or removable computer-readable storage media, including a non-transitory computer readable medium, magnetic computer disks, flash drives, CD-ROM, in-memory device or other suitable media to receive output from and provide input to supply chain network **100**.

[0023] Computer **150** may include one or more processors **156** and associated memory to execute instructions and manipulate information according to the operation of supply chain network **100** and any of the methods described herein. One or more processors **156** may execute an operating

system program stored in memory to control the overall operation of computer **150**. For example, one or more processors **156** control the reception and transmission of signals within the system. One or more processors **156** execute other processes and programs resident in memory, such as, for example, registration, identification or communication and moves data into or out of the memory, as required by an executing process. In addition, or as an alternative, embodiments contemplate executing the instructions on computer **150** that cause computer **150** to perform functions of the method. Further examples may also include articles of manufacture including tangible computer-readable media that have computer-readable instructions encoded thereon, and the instructions may comprise instructions to perform functions of the methods described herein.

[0024] In addition, and as discussed herein, supply chain network **100** may comprise a cloud-based computing system having processing and storage devices at one or more locations, local to, or remote from assortment planner **110**, inventory system **120**, transportation network **130**, and one or more supply chain entities **140**. In addition, each of one or more computers **150** may be a work station, personal computer (PC), network computer, notebook computer, tablet, personal digital assistant (PDA), cell phone, telephone, smartphone, wireless data port, augmented or virtual reality headset, or any other suitable computing device. In an embodiment, one or more users may be associated with assortment planner **110**, inventory system **120**, transportation network **130**, and one or more supply chain entities **140**.

[0025] These one or more users may include, for example, a “manager” or a “planner” handling generation of planograms and instruction sets, managing the inventory of items, imaging items, managing storage and shipment of items, and/or one or more related tasks within supply chain network **100**. In addition, or as an alternative, these one or more users within supply chain network **100** may include, for example, one or more computers **150** programmed to autonomously handle, among other things, planogram planning, shelving resets, task management, communication and assignment of planograms and instructions, issue identification and resolution, controlling manufacturing equipment, and adjusting various levels of manufacturing and inventory levels at various stocking points and distribution centers **146**, and/or one or more related tasks within supply chain network **100**.

[0026] One or more supply chain entities **140** represent one or more supply chain network **100**s, including one or more enterprises, such as, for example networks of one or more suppliers **142**, manufacturers **144**, distribution centers **146**, retailers **148** (including brick and mortar and online stores), customers, and/or the like. Each of one or more supply chain entities **140** may comprise one or more cameras, depths sensors, infrared imaging devices, and/or other data-gathering components to capture images, depth data, and/or other data regarding one or more products and product dimensions delivered to each supply chain entity. Suppliers **142** may be any suitable entity that offers to sell or otherwise provides one or more items (i.e., materials, components, or products) to one or more manufacturers **144**. Suppliers **142** may comprise automated distribution systems **143** that automatically transport products to one or more manufacturers **144** based, at least in part, on planograms and instruction sets determined by assortment planner **110** and/or one or more other factors described herein.

[0027] Manufacturers **144** may be any suitable entity that manufactures at least one product. Manufacturers **144** may use one or more items during the manufacturing process to produce any manufactured, fabricated, assembled, or otherwise processed item, material, component, good, or product. In one embodiment, a product represents an item ready to be supplied to, for example, one or more supply chain entities **140** in supply chain network **100**, such as retailers **148**, an item that needs further processing, or any other item. Manufacturers **144** may, for example, produce and sell a product to suppliers **142**, other manufacturers **144**, distribution centers **146**, retailers **148**, a customer, or any other suitable person or entity. Manufacturers **144** may comprise automated robotic production machinery **145** that produce products based, at least in part, on planograms and instruction sets determined by assortment planner **110** and/or one or more other factors described

herein.

[0028] Distribution centers **146** may be any suitable entity that offers to store or otherwise distribute at least one product to one or more retailers **148** and/or customers. Distribution centers **146** may, for example, receive a product from a first one or more supply chain entities **140** in supply chain network **100** and store and transport the product for a second one or more supply chain entities **140**. Distribution centers **146** may comprise automated warehousing systems **147** that automatically remove products from and place products into inventory based, at least in part, on one or more planograms generated by assortment planner **110**.

[0029] Retailers **148** may be any suitable entity that obtains one or more products to sell to one or more customers. Retailers **148** may comprise a corporate structure having a retail headquarters and one or more retail stores. Retail headquarters comprises a central planning office with oversight of one or more retail stores. Retailer stores may comprise any online or brick-and-mortar store, including stores with shelving systems **149**. The one or more retail stores may sell products according to rules, strategies, orders, and/or guidelines developed by one or more retail headquarters. For example, retail headquarters may create planograms that determine how the store will shelf or display one or more products. Although planogram execution may be performed by a retail employee, embodiments contemplate automated configuration of shelving and retail displays. This may include, for example, automated robotic shelving machinery that places products on shelves or automated shelving that automatically adjusts based, at least in part, on the planograms. Shelving systems **149** may comprise, for example, various racks, fixtures, brackets, notches, grooves, slots, or other attachment devices for fixing shelves in various configurations. These configurations may comprise shelving with adjustable lengths, heights, and other arrangements.

[0030] Although one or more supply chain entities **140** are illustrated and described as separate and distinct entities, the same entity may simultaneously act as any one of one or more supply chain entities **140**. For example, one or more supply chain entities **140** acting as a manufacturer can produce a product, and the same one or more supply chain entities **140** can act as a supplier to supply an item to itself or another one or more supply chain entities **140**. Although one example of supply chain network **100** is illustrated and described in FIG. 1, embodiments contemplate any configuration of supply chain network **100**, without departing from the scope described herein.

[0031] In one embodiment, each of assortment planner **110**, inventory system **120**, transportation network **130**, computer **150**, and supply chain entities **140** may be coupled with network **160** using communication links **162-170**, which may be any wireline, wireless, or other link suitable to support data communications between assortment planner **110** and network **160** during operation of supply chain network **100**. Although communication links **162-170** are shown as generally coupling assortment planner **110**, inventory system **120**, transportation network **130**, one or more supply chain entities **140**, and computer **150** to network **160**, any of assortment planner **110**, inventory system **120**, transportation network **130**, one or more supply chain entities **140**, and computer **150** may communicate directly with each other, according to particular needs.

[0032] In another embodiment, network **160** includes the Internet and any appropriate local area networks (LANs), metropolitan area networks (MANs), or wide area networks (WANs) coupling assortment planner **110**, inventory system **120**, transportation network **130**, one or more supply chain entities **140**, and computer **150**. For example, data may be maintained locally to, or externally of, assortment planner **110**, inventory system **120**, transportation network **130**, one or more supply chain entities **140**, and computer **150** and made available to one or more associated users of assortment planner **110**, inventory system **120**, transportation network **130**, one or more supply chain entities **140**, and computer **150** using network **160** or in any other appropriate manner. For example, data may be maintained in a cloud database at one or more locations external to assortment planner **110**, inventory system **120**, transportation network **130**, one or more supply chain entities **140**, and computer **150** and made available to one or more associated users of assortment planner **110**, inventory system **120**, transportation network **130**, one or more supply

chain entities **140**, and computer **150** using the cloud or in any other appropriate manner. Those skilled in the art will recognize that the complete structure and operation of network **160** and other components within supply chain network **100** are not depicted or described. Embodiments may be employed in conjunction with known communications networks and other components.

[0033] In accordance with the principles of embodiments described herein, assortment planner **110** may generate planograms and instruction sets for the inventory of one or more supply chain entities **140** in supply chain network **100**. Furthermore, assortment planner **110**, inventory system **120**, and/or transportation network **130** may instruct automated machinery (i.e., robotic warehouse systems, robotic inventory systems, automated guided vehicles, mobile racking units, automated robotic production machinery **145**, robotic devices and the like) to adjust product mix ratios, inventory levels at various stocking points, production of products of manufacturing equipment, proportional or alternative sourcing of one or more supply chain entities **140**, the configuration and quantity of packaging and shipping of products, and the display of products at one or more retail locations based on one or more planograms and instruction sets, generated plans and policies and/or current inventory or production levels. When the inventory of an item falls to a reorder point, assortment planner **110** may then automatically adjust product mix ratios, inventory levels, production of products of manufacturing equipment, and proportional or alternative sourcing of one or more supply chain entities **140** until the inventory is resupplied to a target quantity.

[0034] For example, the methods described herein may include computers **150** receiving product data from automated machinery having at least one sensor and the product data corresponding to an item detected by the sensor of the automated machinery. The received product data may include an image of the item, an identifier, as described above, and/or other product data associated with the item (dimensions, texture, estimated weight, and any other like data). The method may further include computers **150** looking up the received product data in database system associated with assortment planner **110**, inventory system **120**, and/or transportation network **130** to identify the item corresponding to the product data received from the automated machinery.

[0035] Computers **150** may also receive, from the automated machinery, a current location of the identified item. Based on the identification of the item, computers **150** may also identify (or alternatively generate) a first mapping in database system, where the first mapping is associated with the current location of the item. Computers **150** may also identify a second mapping in database system, where the second mapping is associated with a past location of the identified item. Computers **150** may also compare the first mapping and the second mapping to determine if the current location of the identified item in the first mapping is different than the past location of the identified item in the second mapping. Computers **150** may then send instructions to the automated machinery based, at least in part, on one or more differences between the first mapping and the second mapping such as, for example, to locate an item to add to or remove from a shelf or an inventory of or shipment for one or more supply chain entities **140**. In addition, or as an alternative, assortment planner **110** monitors the supply chain constraints of one or more items at one or more supply chain entities **140** and adjusts the orders and/or inventory of one or more supply chain entities **140** based on the supply chain constraints.

[0036] FIG. 2 illustrates assortment planner **110** of FIG. 1 in greater detail in accordance with an embodiment. As discussed above, assortment planner **110** may comprise one or more computers **150** at one or more locations including associated input devices **152**, output devices **154**, non-transitory computer-readable storage media, processors **156**, memory, or other components for receiving, processing, storing, and communicating information according to the operation of supply chain network **100**. Additionally, assortment planner **110** comprises server **112** and database **114**. Although assortment planner **110** is shown as comprising a single server **112** and a single database **114**, embodiments contemplate any suitable number of computers **150**, servers **112**, or databases **114** internal to or externally coupled with assortment planner **110**. According to some embodiments, assortment planner **110** may be located internal to one or more retailers **148** of one

or more supply chain entities **140**. In other embodiments, assortment planner **110** may be located external to one or more retailers **148** of one or more supply chain entities **140** and may be located in for example, a corporate retailer of the one or more retailers **148**, according to particular needs. [0037] Server **112** may comprise user action processor **202**, visualization engine **204**, aggregation engine **206**, and hindsight engine **208**. Although server **112** is illustrated and described as comprising a single user action processor **202**, visualization engine **204**, aggregation engine **206**, and hindsight engine **208**, embodiments contemplate any suitable number or combination of these located at one or more locations, local to, or remote from assortment planner **110**, such as on multiple servers **112** or computers **150** at any location in supply chain network **100**.

[0038] User action processor **202** may receive and process user input, such as, for example, input received by input device **152** of one or more computers **150**, including but not limited to a keyboard or mouse. One or more computers **150** may transmit input to assortment planner **110** using one or more communication links **162-170**. User action processor **202** may register the input from one or more computers **150** and transmit the input to visualization engine **204** and/or other assortment planner **110** modules to instigate or control module functions.

[0039] Visualization engine **204** generates one or more graphical user interface (GUI) displays for display on one or more computer monitors and/or other output devices **154**. According to embodiments, visualization engine **204** may access database **114**, including but not limited to historical data **220**, product attributes data **222**, product sales history data **224**, planogram dimension data **228**, metrics data **230**, aggregated data **232**, classification thresholds data **234**, classified data **236**, individual attribute strength data **238**, filtered data **240**, and multi-attribute strength data **242**, and may generate one or more graphical user interface displays to display the data. The one or more graphical user interface displays may convey information, including recommendations, product dimensions, sales data, KPI data, and/or any other type of information about supply chain network **100** and the ongoing sales of products sold throughout supply chain network **100**. In an embodiment, visualization engine **204** may process input transmitted by user action processor **202**, and, in response to the input, may generate one or more subsequent graphical user input displays that allow one or more computers **150** to interact with the graphical user interface displays and to review different information displayed by visualization engine **204**.

[0040] Aggregation engine **206** may aggregate product-store combinations according to selected metrics and may identify the strength of individual attributes, and save such product-store combinations in aggregated data **232**. Aggregation engine **206** may also classify high-performing and low-performing product-store combinations within aggregated data **232** and save such classified data as classified data **236**. Aggregation engine **206** may use classification thresholds data **234** in order to classify the product-store combinations.

[0041] Hindsight engine **208** may identify the strength of individual attributes within classified data **236** and save such attribute strength as individual attribute strength data **238**. Hindsight engine **208** may also filter the attributes stored in individual attribute strength data **238** according to the identified strength of the individual attributes, and save the filtered attributes as filtered data **240**. For example, hindsight engine **208** may filter all attributes falling below a particular threshold count, such as if a particular attribute is associated with only **1** product-store combination, though other filter thresholds may be used. Hindsight engine **208** may also estimate multi-combination strengths for attributes, as described in greater detail below, and save the multi-combination strengths for attributes as multi-attribute strength data **242**.

[0042] Database **114** may comprise one or more databases **114** or other data storage arrangements at one or more locations, local to, or remote from, server **112**. Database **114** comprises, for example, historical data **220**, including but not limited to product placement data **222**, product attributes data **224**, product sales history data **226**, and planogram dimension data **228**; metrics data **230**; aggregated data **232**; classification thresholds data **234**; classified data **236**; individual attribute strength data **238**; filtered data **240**; and multi-attribute strength data **242**. Although

database **114** is illustrated and described as comprising historical data **220**, product placement data **222**, product attributes data **224**, product sales history data **226**, planogram dimension data **228**, metrics data **230**, aggregated data **232**, classification thresholds data **234**, classified data **236**, individual attribute strength data **238**, filtered data **240**, and multi-attribute strength data **242**, embodiments contemplate any suitable number or combination of these, located at one or more locations, local to, or remote from, assortment planner **110** according to particular needs.

[0043] Historical data **220** comprises, for example, any data relating to past sales, past demand, purchase data, promotions, events, or the like of one or more products and/or one or more supply chain entities **140**. Historical data **220** may be stored at time intervals such as, for example, by the minute, hour, daily, weekly, monthly, quarterly, yearly, or any suitable time interval, including substantially in real time.

[0044] Product placement data **222** may comprise placement data and category data for each of the products stored in product attributes data **224**. Product categories may organize products into any number of separate or overlapping categories with one or more placement constraints. By way of example only and not by way of limitation, product placement data **222** may comprise, for a series of drink products, alcoholic and non-alcoholic categories, beer, wine, and spirits categories, perishable and nonperishable categories, and/or any other categories applicable to the products, along with placement requirements for each of the categories of products. Although particular product placement data **222** and categories are described, embodiment contemplate any product placement data **222** and categories, according to particular needs.

[0045] Product attributes data **224** may comprise product identifiers, item identifiers, attribute data (including, for example, brands, dimensions, product images, three-dimensional product representations, and the like), inventory levels, and other like data about one or more products, product hierarchies, substitutable products, target service levels, and the like. Product attributes data **224** may comprise (X,Y) locations of each and every product in the planogram with reference to a left-bottom corner.

[0046] Product sales history data **226** comprises data relating to the sales effects each product may have when displayed next to other products on one or more shelves. Product sales history data **226** may comprise, for example, space elasticity data for a particular product and a particular retail location, and/or the increases or decreases a particular product may have on other products displayed next to the product on a retail shelf. According to embodiments, product sales history data **226** may comprise any one or more general cost or sales price components of one or more items in supply chain network **100**, such as, for example, holding costs, batch costs, stockout costs, inventory costs, order costs, backorder costs, and backlog costs, and/or sale prices for one or more items. Although particular product sales history data **226** are described, embodiment contemplate any type of product sales history data **226**, according to particular needs.

[0047] Planogram dimension data **228** may comprise data relating to one or more planograms. According to some embodiments, planograms comprise, among other data, three-dimensional capacity data of each shelf along with the fixture dimensions and cubic capacity of overall available space, and images of products arranged in a similar layout that the products will appear on the shelves of a retail store. According to other embodiments, planograms comprise one or more digital formats that represent products and product placement as sequences of data structures that may be more easily transmitted and modified by modules or components of one or more of assortment planner **110**, one or more computers **150**, and/or one or more retailers **148**, businesses, or supply chain entities **140** in a supply chain network **100**. Planogram dimension data **228** may comprise one or more existing and/or currently-active planograms that are in place at the one or more retail locations. In an embodiment, planogram dimension data **228** may include relevant key performance indicators (KPIs) and metrics by which assortment planner **110** measures the performance of one or more products and/or one or more categories that have been displayed at one or more retail locations according to one or more planograms.

[0048] Metrics data **230** may store data related to one or more metrics used to measure the performance of one or more products, stores, and/or attributes. Example metrics include revenue, profit, inventory turnover, or any other metric according to the preference of a supply chain planner or other user of assortment planner **110**.

[0049] Aggregated data **232** stores data related to one or more aggregated product-store combinations. In embodiments, aggregated data **232** may be a set of product-store combinations associated with a metric as specified in metrics data **230**. For example, if the metric selected is revenue, aggregated data **232** comprises a set of each product-store combination within the supply chain network, or a selected subset of the supply chain network, along with the revenue for each product-store combination over a particular time period, such as a previous year, a previous month or the like.

[0050] Classification thresholds data **234** may store data specifying the qualities, elements, or transition thresholds between classifications of products (including, but not limited to, classifying products as “high-performing” and “low-performing”). For example, one classification threshold may be 50%, that is, a top half of product-store combinations, according to the metric selected, will be classified as “high-performing” while the corresponding bottom half will be classified as “low-performing.” Other classification thresholds, such as 80% (meaning only the top 20% is classified as high-performing) or 20% (meaning the top 80% is classified as high-performing) may be used according to particular needs or preferences.

[0051] Classified data **236** may store data related to classified product-store combinations. In embodiments, classified data **236** may contain the same product-store combinations as aggregated data **232**, classified to specify which product-store combinations are high-performing and which product-store combinations are low-performing.

[0052] Individual attribute strength data **238** may store data related to quantifying or estimating the individual strength of one or more attributes. In embodiments, individual attribute strength data **238** comprises the estimated likelihood that a particular attribute results in a product-store combination being classified as high-performing. For example, for a set of apparel products, if 90% of products with a fabric color of red have been classified as high-performing, then the attribute “red” may have an individual attribute strength of 0.9. In other examples, attributes may instead be a class of like-attributes, such as “red shades,” “basic colors,” or “premium colors,” rather than a particular individual attribute, according to particular needs and preference.

[0053] Filtered data **240** may store data relating to the attributes whose individual attribute strengths meet or exceed one or more specified thresholds, as described below. In embodiments, if a particular attribute is associated with too few product-store combinations, such an attribute may be filtered from individual attribute strength data **238**. Such filtering action reduces the reliance on small sample sizes, and the potential for outlier or anomalous data to introduce noise into the assortment analysis.

[0054] Multi-attribute strength data **242** may store data related to the multi-combination strengths for potential combinations of attributes. In embodiments, hindsight engine **208** may use filtered data **240** to generate multi-attribute strengths by combining the filtered attribute strengths saved in filtered data **240**. For example, if an attribute “red” has an individual attribute strength of 0.9, and a second attribute “short sleeve” has an individual attribute strength of 0.7, then hindsight engine **208** may determine that the multi-attribute “red and short sleeve” has a multi-attribute strength of 0.8, though the exact computation of multi-attribute strength may not be a mean or average calculation between individual attribute strengths.

[0055] In addition, or as an alternative, embodiments may use location data and location attributes to derive one or more insights, hindsight, and/or other recommendations/associations generated by assortment planner **110**. For example, store type cold region and material cotton might have good strength, such that, the final analysis may also include location attributes along with product attributes to arrive at final insights, hindsight, and/or other recommendations.

[0056] FIG. 3 illustrates an exemplary method **300** of assortment planner **110** extracting hindsights from assortment planning data, according to an embodiment. According to embodiments, method **300** of FIG. 3 may change filter attributes according to individual strength of the filter attributes based on the individual strength and/or the frequency of occurrence of the combination. The following method **300** proceeds by one or more activities, which although described in a particular order may be performed in one or more permutations, combinations, orders, or repetitions, according to particular needs.

[0057] At activity **302**, assortment planner **110** loads historical data **220** into the input data of database **114**. Aggregation engine **206** accesses historical data **220**, which may include product placement data **222**, product attributes data **224**, product sales history data **226**, and/or planogram dimension data **228**, into the input data of database **114**. At activity **304**, user action processor **202** detects input to one or more input devices **152**, and in response to the detected input, user action processor **202** selects one or more metrics with which to aggregate data and classify high-performing and low-performing products. According to embodiments, user action processor **202** may select product margin, product revenue, total profit, and/or any other metric or group of metrics. User action processor **202** stores the selected one or more metrics in metrics data **230** of database **114**.

[0058] At activity **306**, aggregation engine **206** aggregates product-store combinations according to the one or more selected metrics. Aggregation engine **206** accesses the input data and metrics data **230**, and aggregates product-store combinations using the one or more selected metrics. By way of example only and not by way of limitation, in an embodiment in which user action processor **202** selected “margin” as the metric during activity **304** of method **300**, aggregation engine **206** aggregates all products stored in the input data across all stores according to the “margin” key performance indicator (KPI). Aggregation engine **206** stores the aggregated products-store combinations in aggregated data **232**.

[0059] At activity **308**, aggregation engine **206** classifies high-performing and low-performing product-store combinations. Aggregation engine **206** accesses aggregated data **232** and identifies high-performing and low performing products according to classification thresholds stored in classification thresholds data **234**. In an embodiment, aggregation engine **206** may classify all product-store combinations into a binary high-performing/low-performing classification (for example, in an embodiment, “Rockstar” and “Laggard”). In other embodiments, aggregation engine **206** may classify product-store combinations into three classifications (for example, “top performer,” “average performer,” and “problem performer”), or according to any other classification thresholds stored in classification thresholds data **234**. Having classified the product-store combinations, aggregation engine **206** stores the classified product-store combinations in classified data **236**.

[0060] At activity **310**, aggregation engine **206** identifies individual attribute strength. Aggregation engine **206** accesses the classified product-store combinations in classified data **236**. Aggregation engine **206** counts the number of times each attribute for each classified product-store combination appears in each classified category. For example, in an embodiment, the product-store combinations comprise four color attributes **404** (red, yellow, blue, and green) for three shirt style attributes (long-sleeved shirt, T-shirt, and tanktop shirt) sold at ten separate retailers **148** across the United States. In this example, aggregation engine **206** classifies each of the **120** product-store combinations (4 colors attributes×3 shirt style attributes×10 separate retailers **148**) according to one of two classifications: Rockstar (indicating a product-store combination sells more units than the average) and Laggard (indicating a product-store combination sells less units than the average). Aggregation engine **206** counts the number of times each attribute (red, yellow, blue, green, etc.) is associated with a Rockstar product-store combination, and the number of times each attribute is associated with a Laggard product-store combination. Aggregation engine **206** stores the counted number of times each attribute is associated with each classification in individual attribute strength

data **238**. For example, in an embodiment, aggregation engine **206** counts that a particular red attribute appears in a Rockstar product-store combination 90% of the time and in a Laggard product-store combination 10% of the time, and stores an individual attribute strength of 0.9 for the red attribute in individual attribute strength data **238**.

[0061] At activity **312**, hindsight engine **208** filters the attributes according to frequency of the attribute combination that appears in the data. As an example only and not by way of limitation, and as shown in TABLE 1 red, sleeveless combination is not passed to next step, since frequency of the combination is less than threshold of 100. In this way, the insight derived from the data is only valid, if that combinations appears minimum number of times.

TABLE-US-00001 TABLE 1 Frequency Filtered and passed Combination Probability (threshold:100) to next step Red, 0.75 298 Yes cotton Red, 0.88 35 No, since frequency Sleeveless is lower than threshold Pink, 0.21 150 Yes full sleeve

[0062] At activity **314**, hindsight engine **208** estimates multi-combination strengths for attributes. Hindsight engine **208** accesses aggregated data **232**, classified data **236**, individual attribute strength data **238**, and filtered data **240**. Hindsight engine **208** uses the aggregated products-store combinations stored in aggregated data **232**, the classified product-store combinations stored in classified data **236**, and the counted number of times each attribute is associated with each classification in individual attribute strength data **238** to estimate multi-combination strengths for potential combinations of attributes, using only the attributes stored in filtered data **240** that met or exceeded the filtering threshold described above in association with activity **312**.

[0063] In an embodiment, hindsight engine **208** uses association rule mining to estimate multi-combination strengths for potential combinations of attributes. Hindsight engine **208** may use a multi-level attribute strength analysis during association rule mining to estimate multi-combination strengths for potential combinations of attributes. Hindsight engine **208** may construct a network of nodes and edges, an example of which is illustrated by FIG. 5, in which each node represents an attribute (such as, for example, “Sleeveless” or “Round”) and edges connecting nodes represent probabilities that the strength of a particular attribute was classified as high-performing or low-performing. Hindsight engine **208** may traverse the network of nodes and edges to identify attribute combinations that satisfy the one or more metrics in combination (for example, in an embodiment, “Red” and “Sleeveless” together may be associated with high-performing product-store combinations 90% of the time, but “Red” as a single attribute may be associated with high-performing product-store combinations only 34% of the time, and “Sleeveless” only 15% of the time). Having estimated the multi-combination strengths for potential combinations of attributes, hindsight engine **208** may store the multi-combination strengths for potential combinations of attributes in multi-attribute strength data **242**.

[0064] At activity **316**, assortment planner **110** determines whether to return to activity **312**, where hindsight engine **208** may further filter the remaining attributes according to one or more new specified thresholds, or to move to activity **318** of method **300**, described below. In an embodiment, user action processor **202** may react to input to one or more input devices **152**, directing assortment planner **110** to return to activity **312** and to specify, in response to input device **152** input, one or more new specified thresholds by which to narrow the attributes. In other embodiments, assortment planner **110** may repeatedly execute activity **312** and activity **314** of method **300**, setting one or more new specified thresholds every time it executes activity **312**, until assortment planner **110** has generated a specified number of estimated multi-combination strengths for attributes, or has generated a specified number of insights or has reached a specified termination criteria. If assortment planner **110** detects no instructions to return to activity **312**, or if assortment planner **110** receives instructions to proceed to activity **318**, assortment planner **110** continues to activity **318**.

[0065] At activity **318**, visualization engine **204** displays one or more multi-combination strengths for potential combinations of attributes using one or more GUI displays. Assortment planner **110** then terminates method **300**.

[0066] To illustrate the activities of assortment planner **110** extracting hindsight from assortment planning data, the following example is provided. In this example, assortment planner **110** extract profitability insights and hindsight from a body of sales data related to dresses comprising a plurality of attributes (including color attributes **404** such as “red” and “green”, shape attributes **406** such as “round” and “collared”, and sleeve attributes **408** such as “sleeveless”, “full sleeve” and “half sleeve”) sold at several different retail stores. Although this example illustrates assortment planner **110** executing activities of method **300** in a particular order and configuration, embodiments contemplate assortment planner **110** performing the activities of method **300** in one or more permutations, combinations, orders, or repetitions, according to particular needs.

[0067] In this example, at activity **302**, assortment planner **110** loads dresses historical data **220** into the input data of database **114**. Aggregation engine **206** accesses historical data **220**, which in this example includes product placement data **222**, product attributes data **224**, product sales history data **226**, and/or planogram dimension data **228**, into the input data of database **114**. At activity **304**, user action processor **202** detects input to one or more input devices **152**, and in response to the detected input, user action processor **202** selects the “profitability” metric with which to aggregate data and classify high-performing and low-performing products. User action processor **202** stores the selected profitability metric in metrics data **230** of database **114**.

[0068] Continuing the example, at activity **306**, aggregation engine **206** aggregates dress product-store combinations according to the profitability metrics. Aggregation engine **206** accesses the input data and metrics data **230**, and aggregates product-store combinations using the profitability metric. Aggregation engine **206** stores the aggregated products-store combinations in aggregated data **232**.

[0069] Continuing the example, at activity **308**, aggregation engine **206** classifies high-performing and low-performing product-store combinations for the sale of dresses at different stores according to the profitability metric. Aggregation engine **206** accesses aggregated data **232** and identifies high-performing and low performing products according to classification thresholds stored in classification thresholds data **234**. In this example, aggregation engine **206** classifies all product-store combinations into a binary high-performing (1) or low-performing (0) classification. Having classified the product-store combinations, aggregation engine **206** stores the classified product-store combinations in classified data **236**.

[0070] Continuing the example, at activity **310**, aggregation engine **206** identifies individual attribute strengths. Aggregation engine **206** accesses the classified product-store combinations in classified data **236**. Aggregation engine **206** counts the number of times each attribute for each classified product-store combination appears in each of the high-performing (1) and low-performing (0) categories. Aggregation engine **206** stores the counted number of times each attribute is associated with each classification in individual attribute strength data **238**.

Visualization engine **204** accesses individual attribute strength data **238** and generates individual attribute strength data GUI display **400**, illustrated by FIG. **4**.

[0071] FIG. **4** illustrates individual attribute strength data GUI display **400**, according to an embodiment. In an embodiment, individual attribute strength data GUI display **400** may display products according to product ID **402** and associated attributes **404-408** (color attributes **404**, shape attributes **406**, sleeve attributes **408**), as well as whether that particular product/attribute combination **410a-410i** and/or product-store combination was classified as high-performing or low-performing. Although FIG. **4** illustrates individual attribute strength data GUI display **400** in a particular configuration, embodiments contemplate visualization engine **204** generating individual attribute strength data GUI displays **400** of any configuration and displaying any database data, according to particular needs.

[0072] Continuing the example, at activity **312**, hindsight engine **208** filters the attributes according to individual attribute strength. Hindsight engine **208** accesses individual attribute strength data **238** and multi-attribute strength data **242**, and filters all attributes that have frequency below a specified

threshold out of the hindsight extraction method. Hindsight engine **208** filters out all attributes with combination frequencies of **100** or lower as shown in Table 1. Hindsight engine **208** stores the attributes whose individual attribute combination frequencies are above the specified threshold in filtered data **240**.

[0073] Continuing the example, at activity **314**, hindsight engine **208** estimates multi-combination strengths for attributes. Hindsight engine **208** accesses aggregated data **232**, classified data **236**, individual attribute strength data **238**, and filtered data **240**. Hindsight engine **208** uses the aggregated products-store combinations stored in aggregated data **232**, the classified product-store combinations stored in classified data **236**, and the counted number of times each attribute is associated with each classification in individual attribute strength data **238** to estimate multi-combination strengths for potential combinations of attributes, using only the attributes stored in filtered data **240** that met or exceeded the filtering threshold described above in association with activity **312**. In this example, hindsight engine **208** uses association rule mining to estimate multi-combination strengths for potential combinations of attributes. Hindsight engine **208** constructs a network of nodes and edges, and traverses the network of nodes and edges to identify attribute combinations that best satisfy the profitability metric. Having estimated the multi-combination strengths for potential combinations of attributes, hindsight engine **208** may store the multi-combination strengths for potential combinations of attributes in multi-attribute strength data **242**. Visualization engine **204** accesses multi-attribute strength data **242** and generates multi-attribute strength data GUI display **500**, illustrated by FIG. 5.

[0074] FIG. 5 illustrates multi-attribute strength data GUI display **500**, according to an embodiment. In an embodiment, multi-attribute strength data GUI display **500** may display products according to product attributes and whether particular products and/or attributes are more likely or less likely to correspond to high-performing and low-performing product-store combinations. Multi-attribute strength data GUI display **500** comprises single attribute strength chart **510** and multi-attribute strength chart **514**. Although FIG. 5 illustrates multi-attribute strength data GUI display **500** in a particular configuration, embodiments contemplate visualization engine **204** generating multi-attribute strength data GUI displays **500** of any configuration and displaying any database data, according to particular needs.

[0075] Single attribute strength chart **510** shows a set of single attribute nodes, each with an individual attribute strength, arranged in a graph. In this embodiment, single attribute strength chart **510** comprises the nodes “sleeveless,” “round,” “half sleeve,” “collared,” “full sleeve,” and “Chinese.” Single attribute strength chart **510** further comprises edges **512A** and **512B**. Edges **512A** and **512B** may be color-coded according to the strength of each edge **512**. Edge **512A** comprises a multi-attribute strength for the attributes “sleeveless” and “round,” which in this example is 0.9. Edge **512B** comprises a multi-attribute strength for the attributes “round” and “half sleeve,” which in this example is 0.6.

[0076] Multi-attribute strength chart **514** shows a set of attribute nodes, arranged in a graph. In this embodiment, multi-attribute strength chart **510** comprises multi-attribute node **516A** and **516B** “sleeveless, round,” and “round, half sleeve,” as well as single attribute node **518** “red.” Edges **512A** and **512B** of single attribute strength chart **510** are used as nodes in multi-attribute strength chart **514**. Multi-attribute node **516A** corresponds to edge **512A** of single attribute strength chart **510**, and multi-attribute node **516B** corresponds to edge **512B** of single attribute strength chart **510**. In other embodiments, multi-attribute strength charts may be further condensed, using the edges of multi-attribute strength chart **514** to generate nodes for a second multi-attribute strength chart.

[0077] Continuing the example, at activity **316**, assortment planner **110** determines whether to return to activity **312**, where hindsight engine **208** may further filter the remaining attributes according to one or more new specified thresholds, or to move to activity **318** of method **300**, described below. In this example, assortment planner **110** reacts to input to input device **152** instructing assortment planner **110** to proceed to activity **318**.

[0078] Continuing the example, at activity **318**, visualization engine **204** displays one or more multi-combination strengths for potential combinations of attributes using one or more GUI displays. In this example, visualization engine **204** generates strength, weakness, opportunity and threat (SWOT) analysis GUI display **600**, illustrated by FIG. **6**, and recommendation GUI display **700**, illustrated by FIG. **7**. Assortment planner **110** then terminates method **300**.

[0079] FIG. **6** illustrates SWOT analysis GUI display **600**, according to an embodiment. In an embodiment, SWOT analysis GUI display **600** may display data stored in individual attribute strength data **238**, filtered data **240**, multi-attribute strength data **242**, and/or other data related to one or more insights, hindsights, and/or other recommendations/associations generated by assortment planner **110**. In an embodiment, assortment planner **110** may identify strengths **602**, weaknesses **604**, opportunities **606**, and trends **608** using individual attribute strength data **238** (i.e., probabilities **610** and frequencies **612**), filtered data **240**, multi-attribute strength data **242**, and/or other database data, and visualization engine **204** may display the strengths **602**, weaknesses **604**, opportunities **606**, and trends **608** using one or more SWOT analysis GUI displays **600**. Although FIG. **6** illustrates SWOT analysis GUI display **600** in a particular configuration, embodiments contemplate visualization engine **204** generating SWOT analysis GUI displays **600** of any configuration and displaying any database data, according to particular needs.

[0080] FIG. **7** illustrates second method **700** by which hindsight engine **208** may estimate the multi-combination strength of attributes, according to embodiments. Although FIG. **8** illustrates second method **700** in a particular configuration, embodiments contemplate assortment planner **110** executing activities in any order estimating the multi-combination strength of attributes in any configuration or permutation, according to particular needs.

[0081] In an embodiment, hindsight engine **208** aggregates item-store data at activity **702**. The result of activity **702** is illustrated by aggregation chart **702A**. Aggregation chart **702A** shows an example with six product-store combinations, and the associated revenues of the six product-store combinations.

[0082] At second activity **704**, hindsight engine **208** classifies high-performing and low-performing combinations. The result of activity **704** is illustrated by classification chart **704A**, which separates the product-store combinations into high-performing and low-performing combinations according to the revenue of each product-store combination. In embodiments, the entries of classification chart **704A** may be color coded, or otherwise marked, to indicate which combinations are high-performing and which combinations are low-performing, as also indicated by the “Classify” column of classification chart **704A**.

[0083] At third activity **706**, hindsight engine **208** derives attribute strengths from combinations. The results of activity **706** are illustrated by individual attribute chart **706A**. Individual attribute chart **706A** shows the probability of success (as defined by the likelihood of a particular attribute being associated with a high-performing combination, for 5 attributes).

[0084] At fourth activity **708**, hindsight engine **208** estimates multi-combination strength. The results of activity **708** are illustrated in multi-attribute chart **708A**. Multi-attribute chart **708A** shows the probability of success for combinations of the attributes shown in single-attribute chart **706A**. In this example, 3 multi-attributes are illustrated, when in fact according to all possible combinations of the 5 attributes shown in **706A** would result in a larger number of multi-attributes.

[0085] FIG. **8** illustrates insight analysis GUI display **800**, according to an embodiment. In an embodiment, visualization engine **204** may access data stored in database **114**, including but not limited to historical data **220**, aggregated data **232**, classified data **236**, individual attribute strength data **238**, and multi-attribute strength data **242**, and may generate one or more insight analysis GUI displays **800** to illustrate interactions among different attributes. Insight analysis GUI display **800** comprises insight chart **810** and insight graph **820**. Although FIG. **8** illustrates insight analysis GUI display **800** in a particular configuration, embodiments contemplate visualization engine **204** generating insight analysis GUI displays **800** in any configuration, according to particular needs.

[0086] Insight chart **810** illustrates the relative attribute strengths of nine multi-attributes, which comprise the actual product-store combinations corresponding to three “price” attributes and four “sleeve type” attributes. According to embodiments, visualization engine **204** may color-code attribute interactions, such as, for example, by illustrating potentially profitable collection **812** of attributes with a green color, and illustrating less-profitable collections **814** of attributes using yellow, red, or other colors.

[0087] Insight graph **820** illustrates a graph network of eight single attribute nodes, along with the edges between those nodes. According to embodiments, visualization engine **204** may color-code the edges of insight graph **820**, such as, for example, by illustrating edges with relatively high multi-attribute strengths with a green color, and illustrating edges with relatively low multi-attribute strengths with using yellow, red, or other colors.

[0088] Reference in the foregoing specification to “one embodiment”, “an embodiment”, or “some embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

[0089] While the exemplary embodiments have been illustrated and described, it will be understood that various changes and modifications to the foregoing embodiments may become apparent to those skilled in the art without departing from the spirit and scope of the present invention.

Claims

1. A system for extracting hindsight from assortment planning data, comprising: an assortment planner comprising a server and configured to: load historical data into input data of a database; detect input to one or more input devices, and in response to the detected input, select one or more metrics with which to aggregate data and classify product performance; aggregate product-store combinations according to the one or more selected metrics; classify the product-store combinations into categories according to classification thresholds; identify individual attribute strength of attributes by counting a number of times each attribute for each classified product-store combination appears in each classified category; filter the attributes according to a frequency of an attribute combination; estimate multi-combination strengths for the attributes; and display one or more multi-combination strengths for potential combinations of the attributes.
2. The system of claim 1, wherein the one or more selected metrics comprise one or more of: product margin, product revenue and total profit.
3. The system of claim 1, wherein the product-store combinations are classified as either high-performing or low-performing.
4. The system of claim 1, wherein the product-store combinations are classified into three classifications.
5. The system of claim 1, wherein the server is further configured to: determine whether to further filter remaining attributes according to one or more new specified thresholds; in response to determining to further filter the remaining attributes, further filter the remaining attributes; and estimate further multi-combination strengths for the remaining attributes.
6. The system of claim 1, wherein the loaded historical data comprises one or more of: product placement data, product attributes data, product sales history data, and planogram dimension data.
7. The system of claim 1, wherein the one or more selected metrics comprise one or more of: product margin, revenue, profit and inventory turnover.
8. A method for extracting hindsight from assortment planning data, comprising: loading, by an assortment planner comprising a server, historical data into input data of a database; detecting, by the server, input to one or more input devices, and in response to the detected input, selecting, by

the server, one or more metrics with which to aggregate data and classify product performance; aggregating, by the server, product-store combinations according to the one or more selected metrics; classifying, by the server, the product-store combinations into categories according to classification thresholds; identifying, by the server, individual attribute strength of attributes by counting a number of times each attribute for each classified product-store combination appears in each classified category; filtering, by the server, the attributes according to a frequency of an attribute combination; estimating, by the server, multi-combination strengths for the attributes; and displaying, by the server, one or more multi-combination strengths for potential combinations of the attributes.

9. The method of claim 8, wherein the one or more selected metrics comprise one or more of: product margin, product revenue and total profit.

10. The method of claim 8, wherein the product-store combinations are classified as either high-performing or low-performing.

11. The method of claim 8, wherein the product-store combinations are classified into three classifications.

12. The method of claim 8, further comprising: determining, by the server, whether to further filter remaining attributes according to one or more new specified thresholds; in response to determining to further filter the remaining attributes, further filtering, by the server, the remaining attributes; and estimating, by the server, further multi-combination strengths for the remaining attributes.

13. The method of wherein the loaded historical data comprises one or more of: product placement data, product attributes data, product sales history data, and planogram dimension data.

14. The method of claim 8, wherein the one or more selected metrics comprise one or more of: product margin, revenue, profit and inventory turnover.

15. A non-transitory computer-readable medium comprising software for extracting hindsights from assortment planning data, the software when executed is configured to: load historical data into input data of a database; detect input to one or more input devices, and in response to the detected input, select one or more metrics with which to aggregate data and classify product performance; aggregate product-store combinations according to the one or more selected metrics; classify the product-store combinations into categories according to classification thresholds; identify individual attribute strength of attributes by counting a number of times each attribute for each classified product-store combination appears in each classified category; filter the attributes according to a frequency of an attribute combination; estimate multi-combination strengths for the attributes; and display one or more multi-combination strengths for potential combinations of the attributes.

16. The non-transitory computer-readable medium of claim 15, wherein the one or more selected metrics comprise one or more of: product margin, product revenue and total profit.

17. The non-transitory computer-readable medium of claim 15, wherein the product-store combinations are classified as either high-performing or low-performing.

18. The non-transitory computer-readable medium of claim 15, wherein the product-store combinations are classified into three classifications.

19. The non-transitory computer-readable medium of claim 15, where the software is further configured to: determine whether to further filter remaining attributes according to one or more new specified thresholds; in response to determining to further filter the remaining attributes, further filter the remaining attributes; and estimate further multi-combination strengths for the remaining attributes.

20. The non-transitory computer-readable medium of claim 15, wherein the loaded historical data comprises one or more of: product placement data, product attributes data, product sales history data, and planogram dimension data.
